



Namal University Mianwali

Department of Computer Science

1. COURSE DETAILS

Course Title:	Natural Language Processing
Course Code:	CSC-355
Credits (Theory + Lab):	3 (3+0)
Lecture Hours/Week:	3
Pre-Requisite:	None

Brief Course Description:

This course introduces Natural Language Processing (NLP), covering the complete NLP pipeline from text preprocessing and linguistic analysis to modern deep learning-based language models. Students study statistical and vector-based text representations, neural network architectures for sequence modeling, and attention-based methods. The course further explores Transformer architecture and pretrained language models such as BERT and GPT, including fine-tuning and alignment techniques. Emphasis is placed on practical implementation, evaluation using standard NLP metrics, ethical considerations, and deployment of NLP systems. A project-based approach enables students to apply modern tools to solve real-world language processing problems.

2. INSTURCTOR DETAILS

Name:	Dr. Muzamil Ahmed
Lecture Timings:	Tuesday (2:00 PM to 3:30 PM) and Wednesday (9:00 AM to 10:30 AM),
Office Hours:	Monday & Thursday (11:00 AM to 2:00 PM)
Office Location:	Office 6
Office Phone:	Ext. 184
Email:	muzamil.ahmed@namal.edu.pk

3. COURSE UNIT WISE MAJOR TOPICS

Unit No.	Topics	No of Teaching Hours
1	Introduction to NLP & Text Processing	6
2	Text Representation & Word Embeddings	7.5
3	Neural Networks & Sequence Modeling	10.5
4	Attention, Transformers & Language Models	7.5
5	Pretrained Models & Core NLP Applications	7.5
6	Evaluation, Ethics & NLP System Development, CCP Project	9
Total Contact Hours		48

4. COURSE LEARNING OUTCOMES (CLOs)

Sr. No	Unit No	CLO Statement	Taxonomy Level	GA
CLO-1	1-2	Explain fundamental concepts, pipelines, and representations used in Natural Language Processing	C2	1
CLO-2	2-3	Apply NLP techniques and machine learning models to solve computational language problems	C3	2
CLO-3	3-5	Design and develop NLP-based solutions using neural networks, Transformers, and pretrained language models for real-world applications	C5	4
CLO-4	2-6	Utilize modern NLP tools, frameworks, pretrained models, and evaluation metrics for building, fine-tuning, and deploying NLP systems	C3	5

5. WEEKLY LECTURE BREAKUP

Lecture #	Unit #	Topics Covered in Class	Reference in Book / Assessments
1	1	Module 1: Intro to NLP and Text Preprocessing Introduction to Course, NLP Applications, Project-Oriented Learning	NLP by Raymond Lee CH-1
2	1	Modern NLP Pipelines: From Text Processing to Language Models	NLP by Raymond Lee CH-1
3	1	Text Preprocessing, Tokenization, Linguistic Basics	NLP with Python by Steven CH-2
4	1	N-gram Language Models	NLP with Python by Steven CH-2
5	2	Module 2: Text Representation Bag-of-Words and TF-IDF, Baseline Models	NLP by Raymond Lee CH-2
6	2	Word2Vec: Skip-Gram and CBOW	NLP by Raymond Lee CH-2
7	2	GloVe and FastText Embeddings	NLP by Raymond Lee CH-2
8	2	Word Vectors and Language Models	Quiz-1 NLP by Raymond Lee CH-2
9	2	Embedding Analysis and Visualization	NLP by Raymond Lee CH-2
10	3	Module 3: Neural Networks for NLP Neural Networks for NLP: Overview	Assignment 1 -Project Milestone 1 Neural Network Methods for NLP by Yoav CH-4
11	3	Feedforward Neural Networks and Backpropagation	Neural Network Methods for NLP by Yoav CH-4
12	3	Dependency Parsing with Neural Networks	Neural Network Methods for NLP by Yoav CH-4
13	3	Sequence Modeling: RNNs	Handbook of NLP by Nitin CH-4
14	3	Advanced Variants of RNNs, Attention	Handbook of NLP by Nitin CH-4

15	3	LSTM and GRU Architectures	Handbook of NLP by Nitin CH-4
16	3	Attention Mechanism in Sequence Models	Assignment 2 -Project Milestone 2 Handbook of NLP by Nitin CH-4
Mid Term Examination			
17	4	Module 4: Transformers Self-Attention: Motivation and Formulation	NLP with Transformers CH-3 by Lewis Tunstall
18	4	Transformer Architecture I: Model Components	NLP with Transformers CH-3 by Lewis Tunstall
19	4	Transformer Architecture-II: Regularization & Optimization	NLP with Transformers CH-3 by Lewis Tunstall
20	4	Encoder–Decoder Transformers	NLP with Transformers CH-4 by Lewis Tunstall
21	4	Language Modeling and Perplexity	NLP with Transformers CH-4 by Lewis Tunstall
22	5	Module 5: Pretrained Language Models Pretraining Objectives: Masked and Causal LMs	Assignment 3 -Project Milestone 3 LLM: A Deep Dive by Uday Kamath CH-2
23	5	Pretrained Language Models: An Overview	LLM: A Deep Dive by Uday Kamath CH-3
24	5	BERT, GPT, and Transfer Learning	LLM: A Deep Dive by Uday Kamath CH-3
25	5	Fine-Tuning Large Language Models	Quiz 2 LLM: A Deep Dive by Uday Kamath CH-6
26	5	Alignment and Post-Training (SFT, RLHF, DPO)	LLM: A Deep Dive by Uday Kamath CH-7
27	6	Module 6: Evaluation and Benchmarking NLP Evaluation Methodologies	Assignment 4 -Project Milestone 4
28	6	Evaluation Metrics: F1, BLEU, ROUGE	Handbook of NLP by Nitin CH-8
29	6	Ethics, Bias, and Responsible NLP	Handbook of NLP by Nitin CH-8
30	6	NLP System Integration and Deployment	Project Work : Project Milestone 5 Handbook of NLP by Nitin CH-8
31	-	Project Presentations – I (Group Project)	
32	-	Project Presentations – II (Group Project)	
Final Term Exam			

Assessment Tool	CLO1	CLO2	CLO3	CLO4
Quizzes	✓	✓		
Assignments		✓	✓	✓
Projects	✓	✓	✓	✓
Mid Term Exam	✓	✓	✓	
Final Term Exam	✓	✓	✓	✓

6. TEACHING MATERIAL

Textbooks:
1. Natural Language Processing by Raymond Lee 1 st Edition 2. Neural Network Methods for Natural Language Processing by Yoav 1 st Edition 3. Large Language Models: A Deep Dive: Bridging Theory and Practice 2024th Edition by Uday Kamath
Reference Material:
1. Handbook of Natural Language Processing (Chapman & Hall/CRC Machine Learning & Pattern Recognition) 2nd Edition

7.	8. POLICY & PROCEDURES						
• Attendance Policy: Students must attend 80% of this course's lectures. Students who fall short of the required attendance percentage cannot appear in the terminal examination. • Plagiarism Policy: Namal University's rules and regulations prohibit plagiarism, copying, and other dishonest behavior. Violators will face serious disciplinary action. • Course Assessment:							
Assessment Tool	Quiz	Assignment	Project/ Tools	CCP	Mid Term Exam	Final Term Exam	Total
Theory (T)	5	15	5	5	30	40	100

9. VERIFICATION
(i) I verify that the content of this document is correct and up-to-date.
Dr. Muzamil Ahmed Instructor's Name and Signature  12-02-2026 Date
(ii) I have reviewed course-outline and state that it complies with Namal University policies and guidelines.
Dr. Muhammad Ali Shahid Name and Signature of Head of Departme  12-02-2026 Date